

Shoban Ahmad

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EDUCATION

• Indian Institute of Technology Delhi

• *B.Tech in Textile and Fiber Engineering*

Hauz Khas, New Delhi

Nov 2022 – May 2026

EXPERIENCE

• Nomadiq Travel (May 2025):

Performed data preprocessing, feature engineering, and model tuning to improve prediction accuracy and generalization. Assisted in Development of flight ticket price prediction model using XGBoost , leveraging temporal, airline, and routebased features and Deployed the trained model, enabling real-time price predictions for unseen routes and dates.

PROJECTS

- **Boolean Retrieval Engine(Aug 2025):** Implemented a Boolean retrieval engine on the CORD-19 dataset, building a custom tokenizer and positional inverted index.Applied gap encoding and variable-length compression, reducing storage by 30 percent while ensuring lossless decompression. Designed a tree-based Boolean query parser with operator precedence and scalable evaluation for efficient query execution.
- **Automated Corpus Analysis and Querying Tool(Data Structures and Algorithms) (Sep 2025):** Addressed the challenge of LLM context-window limits due to token limits by enabling efficient text access for large corporations.Built an inverted index with IDF-based ranking and trie structures to store vocabulary and surface relevant passages.Successfully optimized query answering to $O(n \times f)$ time, where n is question length and f is term frequency in the corpus.
- **Knowledge Graph RAG Q and A Agent (Generative AI and Information Retrieval):** Engineered a Graph RAG system by combining a Neo4j knowledge graph with vector-based retrieval for accurate Q and A.Built a knowledge graph for Wikipedia articles using entity–relation extraction, graph embeddings, and linked passages.Designed an LLM agent that leverages subgraph + text context to deliver grounded, citation-backed answers.
- **Stock Market Price Movement Prediction (Dec 2024):** Collected and Prepared the dataset by processing CSV files containing historical stock prices from Google, implemented a Machine learning Model, Evaluated the model's performance using metrics such as Mean Absolute Error,Root Mean Squared Error(RMSE),and R-Squared.
- **Document Processing Tool (Feb 2024):** Created a search engine that matches input queries with corresponding corpus answers, improving retrieval accuracy. Implemented pattern searching using the KMP algorithm, enhancing the efficiency of the search engine, Successfully achieved $O(m+n)$ time complexity, where m is the length of the pattern and n is the corpus length.

SKILLS

Languages/Libraries:Python, C++, Java, Numpy, Pandas, Matplotlib ,Scikit- Learn,OpenCV, Tensorflow, Keras, PyTorch. **Software Tools:** Jupyter, Github, VS Code, AutoDesk. **Financial Tools:**Financial data analysis, market research, business model evaluation, SQL for structured data analysis, Excel

EXTRA CURRICULAR ACTIVITIES

- **AmplifyME Quant Trading Market Making Competition, conducted by Citadel Securities (Oct 2024):** Competed in simulated trading and market-making challenges, applying quantitative reasoning and buy/sell-side decision-making.Demonstrated strong critical thinking with 83 percent score and decision-making in simulated trading and market-making scenarios.

HONORS AND AWARDS

- Secured All India Rank 12654 in Joint Entrance Examination Advanced amongst 1.2 million candidates, 2022
- Secured 99.16 Percentile in JEE Main Examination amongst 1.2 million candidates.
- Demonstrating a passion for competitive programming with Pupil on Codeforces.

VOLUNTEER EXPERIENCE

- **Representative, Algorithms and Coding Club, IIT Delhi:** Led the club at orientation and tour of stalls. Successfully administered ANCC events with 300+ participants from IIT Delhi.
- **Volunteer, Office of Career Services, IIT Delhi:** Assisted in placement interview processes by managing interview logistics, and assisting recruiters during campus hiring.